# INFO 4220 – Networks II: Market Design
# Spring 2025 – Problem Set 3

**Solution**

## Problem 1 (17 points)

A group of four friends—**Alex, Jordan, Sam, and Riley**—is preparing for a Dungeons & Dragons (D&D) campaign. Each of them has a ranked preference list for the four available classes: **Fighter, Rogue, Wizard, and Cleric**. They each prefer different classes to different extents, and the goal is to assign the classes to have a party in which everyone has a different class.

| **Alex** | **Jordan** | **Sam** | **Riley** |
|---|---|---|---|
| Fighter | Wizard | Rouge | Fighter |
| Rouge | Cleric | Cleric | Wizard |
| Wizard | Fighter | Fighter | Rouge |
| Cleric | Rouge | Wizard | Cleric |

a. Find an ex-ante efficient probabilistic assignment for this problem using the probabilistic serial algorithm using the same eating speed for everyone (4 points)

**Solution:**

Step 1 (time = ½)
- Alex gets ½ fighter.
- Jordan gets ½ Wizard.
- Sam gets ½ Rouge.
- Riley gets ½ Fighter.

**1 point**

Step 2 (time = ¾)
- Alex gets ¼ Rouge
- Jordan gets ¼ Wizard.
- Sam gets ¼ Rouge.
- Riley gets ¼ Wizard.

**1 point**

Step 3 (time = 1)

- All get ¼ Cleric.

**1 point**

| | Fighter | Rouge | Wizard | Cleric |
|---|---|---|---|---|
| Alex | ½ | ¼ | 0 | ¼ |
| Jordan | 0 | 0 | ¾ | ¼ |
| Sam | 0 | ¾ | 0 | ¼ |
| Riley | ½ | 0 | ¼ | ¼ |

**1 point**

b. Decompose the result in a) as a lottery over deterministic assignments (5 points)

**Solution**

$$\frac{1}{4}\begin{pmatrix}1&0&0&0\\0&0&1&0\\0&1&0&0\\0&0&0&1\end{pmatrix}+\frac{1}{4}\begin{pmatrix}1&0&0&0\\0&0&0&1\\0&1&0&0\\0&0&1&0\end{pmatrix}+\frac{1}{4}\begin{pmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\\1&0&0&0\end{pmatrix}+\frac{1}{4}\begin{pmatrix}0&0&0&1\\0&0&1&0\\0&1&0&0\\1&0&0&0\end{pmatrix}$$

1 point each matrix, 1 point to have all correct.

c. Is the assignment found in a) ex-post efficient? Motivate the answer and show all the steps you need to draw the conclusion (4 points)

**Solution**

Each of the deterministic assignments is ex-post efficient (0.75 points for showing/explaining each), so the assignment found in a) is ex-post efficient (1 point).

The group decides that Sam will be the Cleric, since she has the most experience playing support characters. Then, the dungeon master (DM) says that it is better to use a different assignment he came up with than the one found in a):

| | **Fighter** | **Wizard** | **Rouge** |
|---|---|---|---|
| **Alex** | 0.5 | 0.3 | 0.2 |
| **Jordan** | 0.3 | 0.3 | 0.4 |

| | | | |
|---|---|---|---|
| **Riley** | 0.25 | 0.5 | 0.25 |

d. Suppose the players' preferences remain unchanged. Find the assignment without Sam and compare it to the one proposed by the DM. Which one is stochastically dominant for each player? Show and exhaustively motivate all the steps you take to draw conclusions. (4 points)

**Solution**

Step 1 (t = ½)
- Alex gets ½ Fighter
- Riley gets ½ Fighter
- Jordan gets ½ Wizard.

Step 2 (t = ¾)
- Alex gets ¼ rouge
- Riley gets ¼ wizard
- Jordan gets ¼ wizard

Step 3 (t = 1)
- All get ¼ rouge

Eating Algorithm Assignment (EA)

| | Fighter | Wizard | Rouge |
|---|---|---|---|
| Alex | ½ | 0 | ½ |
| Riley | ½ | ¼ | ¼ |
| Jordan | 0 | ¾ | ¼ |

**1 point**

We compare the two probabilistic assignments for each.

Alex
- 1st choice: 1/2 in both.
- 1st or 2nd choice: 0.7 in the DM assignment and 1 in the EA assignment.

The EA assignment stochastically dominates the DM assignment.
**1 point**

Riley
- 1st choice: ¼ in DM assignment and ½ in EA assignment.
- 1st or 2nd choice: ¾ in both.

The EA assignment stochastically dominates the DM assignment.
**1 point**

Jordan

- 1st choice: 0.3 in DM assignment and 0.75 in EA assignment.
- 1st or 2nd choice: 0.6 in DM and 1 in EA assignment.

The EA assignment stocastically dominates the DM assignment.
**1 point.**

## Problem 2 (18 points, 3 pts each)

At the start of the semester, the school needs to assign 6 activity rooms (A-F) to 5 student clubs: Boxing, Chorus, Finance, Gaming, Movie. The assignment from last year and their preferences for this year are as below.

| A | B | C | D | E |
|---|---|---|---|---|
| **Boxing** | **Chorus** | **Finance** | **Gaming** | **Movie** |
| C | A | B | E | E |
| B | E | C | A | F |
| A | C | E | D | B |
| F | B | F | C | D |
| D | D | D | F | C |
| E | F | A | B | A |

< Endowment

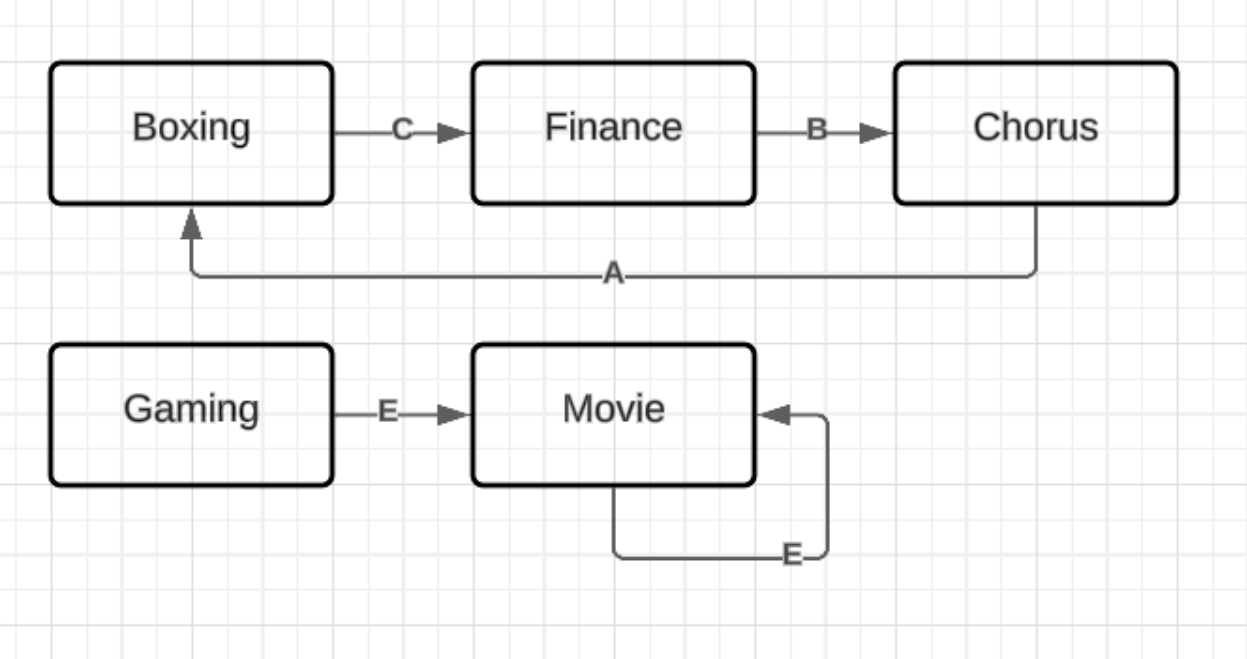
a. One proposal for the assignment is to use serial dictatorship. Use the order of Movie, Gaming, Chorus, Boxing, Finance to find the assignment with serial dictatorship. Is the assignment in the core? Explain why. (3 points)

**Solution:**
Movie – E
Gaming – A
Chorus – C
Boxing – B
Finance – F
**1 point**
It is not in the core because Boxing, Chorus, and Finance would be better by trading between them than participating.
**2 points.**

b. Use TTC to find the assignment. Show all your steps and the result. (4 points)

**Solution**

Step 1:




We assign:

- Boxing – C
- Finance – B
- Chorus – A
- Movie – E

**3 points.**

Step 2

We assign

- Gaming – D

**1 point**

c. Simon says that TTC is better than Serial Dictatorship because TTC generates the only efficient result. Do you agree with him? Explain why. (3 points)

**Solution**

Simon is wrong, because serial dictatorship also generates efficient results (1 point), TTC is better because it is also individually rational and generates assignments in the core (2 points).

Later in the semester, the university announces that Room C will be under renovation next year, meaning the Finance Club will lose its assigned room. Additionally, a new club (VR Club) will be introduced two years later, and Room C will be available again at that time (although

unassigned). To assign rooms to clubs two years later, use the endowment and preferences in the table below.

| A | B | | D | E | | < Endowment |
|---|---|---|---|---|---|---|
| **Boxing** | **Chorus** | **Finance** | **Gaming** | **Movie** | **VR** | |
| C | A | B | E | E | A | |
| B | E | C | A | F | C | |
| A | C | E | D | B | F | |
| F | B | F | C | D | E | |
| D | D | D | F | C | D | |
| E | F | A | B | A | B | |

d. Emma used the Random Serial Dictatorship with Waiting List algorithm and found the assignment: Chorus -> C, Gaming -> D, VR -> F, Movie -> E, Finance -> B, Boxing -> A. Is this assignment efficient? Could you explain to Emma why, or why not, the Random Serial Dictatorship with Waiting List algorithm generates efficient results? (4 points)

**Solution**
It is not efficient, since chorus and boxing could exchange assignments, and they would both be better (2 points).

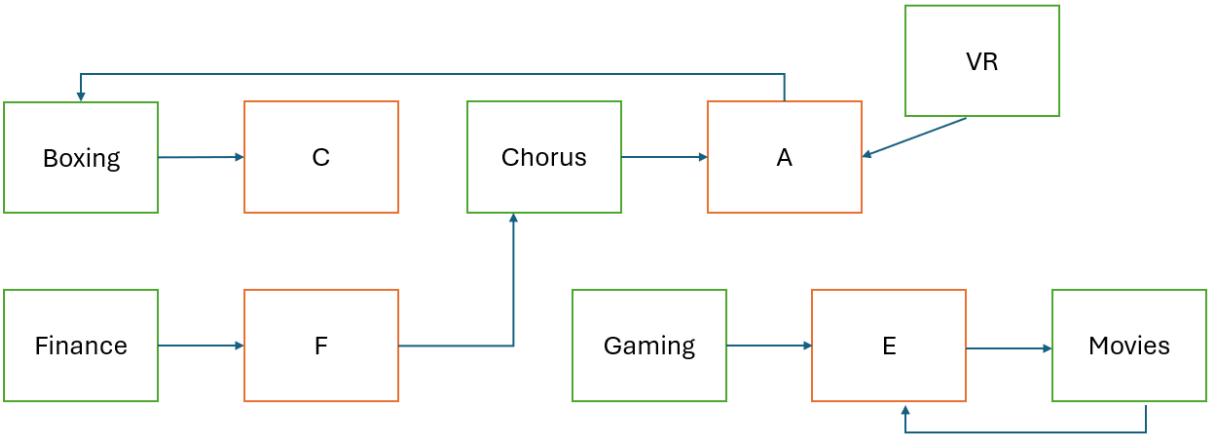
Serial dictatorship with waiting list produces no efficient assignment because occupied items are not available from the beginning. Occupied items only become available after the owners choose something else. This is inefficient if someone who could have chosen the item has to choose before the current owner. (2 points)

e. To find a better assignment, could you recommend one other algorithm to be efficient and strategyproof? Find the assignment with your proposed algorithm. If at any time you need a random order to run the algorithm, use the order: Chorus, Gaming, VR, Movie, Finance, Boxing. (4 points)

**Solution**
TTC or YRMH-IGYT

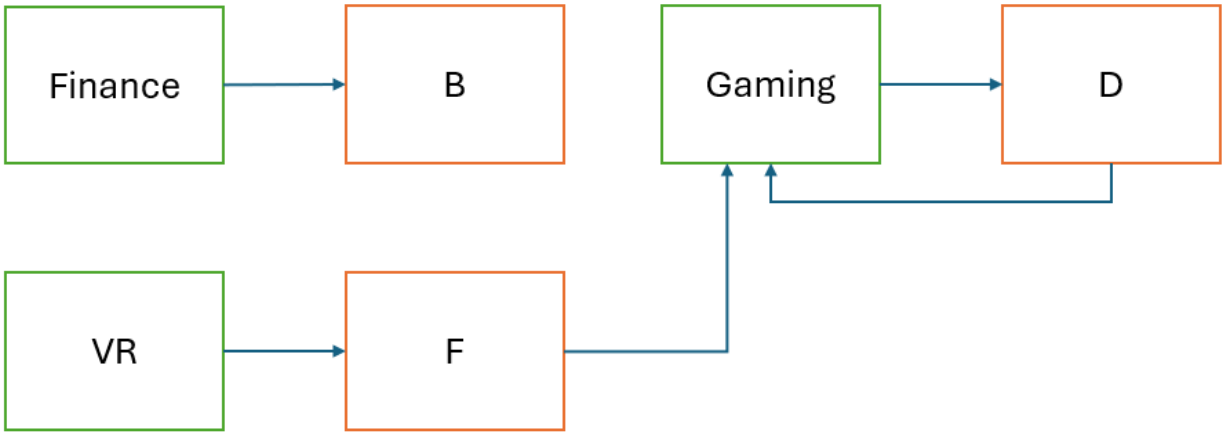
Option 1 – TTC

Step 1:



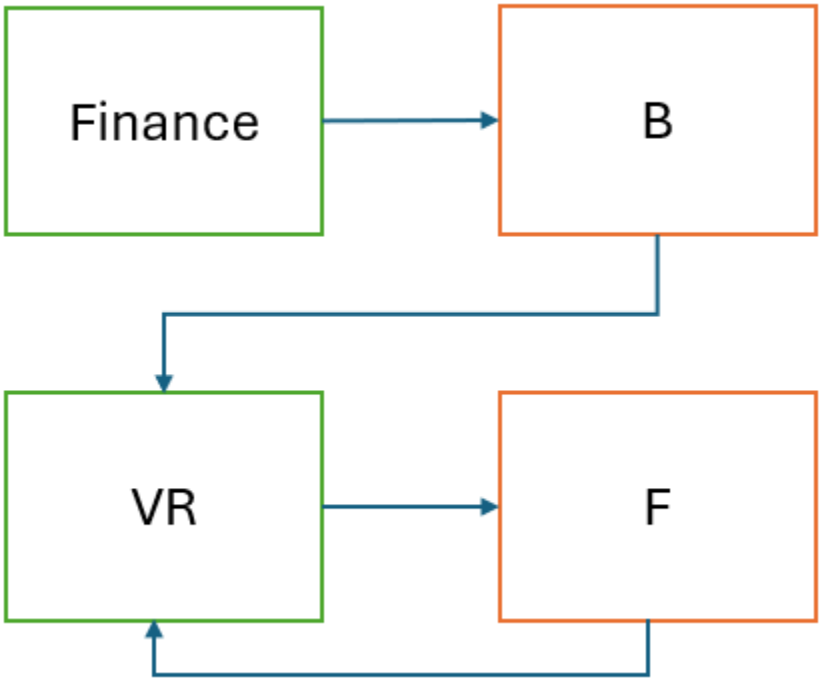

Boxing – C
Chorus – A
Movie – E
**2 points**

Step 2:




Gaming – D
**1 point**

Step 3




VR – F

Step 4
Finance - B

**1 point.**

Option 2: YRMH-IGYT

Order: Chorus, Gaming, VR, Movie, Finance, Boxing

Chorus – A: conflicts with boxing.

**1 point**

Order 2: Boxing, Chorus, Gaming, VR, Movie, Finance

**Boxing – C (leaves A)**
**Chorus – A  (leaves B)**
Gaming  - E: conflicts with Movie.

**2 points**

New order: Movie – Gaming – VR – Finance

**Movie – E**
**Gaming – D**
**VR – F**
**Finance – B**

**1 point**

## Problem 3 (15 points, 5 pts each)

5 students (Alice, Bob, Carol, Denis, Eva) apply to three schools (S1, S2, S3). Each school has strict preferences over students and capacity constraints. The students' true preferences over schools and schools' priority and capacity are as follows:

| **Alice** | **Bob** | **Carol** | **Denis** | **Eva** |
|---|---|---|---|---|
| S2 | S3 | S1 | S2 | S3 |
| S1 | S2 | S3 | S3 | S2 |
| S3 | S1 | S2 | S1 | S1 |

| Capacity | 2 | 1 | 2 |
|---|---|---|---|
| | **S1** | **S2** | **S3** |
| | Carol | Alice | Eva |
| | Alice | Denis | Denis |
| | Bob | Bob | Bob |
| | Denis | Carol | Alice |
| | Eva | Eva | Carol |

a. Run IA to find the school assignment. Show all your steps. Is the result stable? Is it efficient? Justify your answer. What strategies might each student consider for improving their placement when IA is used?

**Solution**

Step 1:
- Alice, Denis → S2 (rejects Denis)
- Bob, Eva → S3
- Carol → S1

**0.5 point**

Step 2
- Denis → S3 (rejected)

Step 3
- Denis → S1

S1 – Denis, Carol
S2 – Alice
S3 – Bob, Eva
**0.5 point**

The assignment is efficient, but it is not stable because there is justified envy. Denis likes S3 better than his assignment and has a strictly higher priority than Bob at S3

**2 points (1 each)**

Strategies
Alice, Bob, Eva and Carol are in their top choices.
Denis could put S3 as his 1st choice to improve assignment.
**2 points**

b. Run DA with students proposing to find the school assignment, assuming students submit their true preferences. Show all your steps. Is the result stable? Is it efficient? Explain why.

**Solution**

Step 1:
- Alice, Denis → S2 (s2 rejects Denis)
- Bob, Eva → S3
- Carol → S1

**1 point**

Step 2:
- Denis → S3, S3 rejects Bob.

**1 point**

Step 3:
- Bob → S2, S2 rejects Bob.

Step 4:
- Bob → S1

S1 – Bob, Carol
S2 – Alice
S3 – Denis, Eva
**1 point**

Efficiency and Stability.
**2 points (1 each)**

c. Suppose that some schools have strong preferences for certain students, so they might not submit their true priority. Would it potentially affect students' welfare if still using DA with student proposing? Justify your answer.

**Solution**

Yes, it would affect (2 points). For example, maybe in this question, S2 prefers Denis over Alice, they could lie about their priority ordering, and Denis would get S2 instead of Alice. Denis would be better and Alice would be worse (3 points for this or a similar explanation).